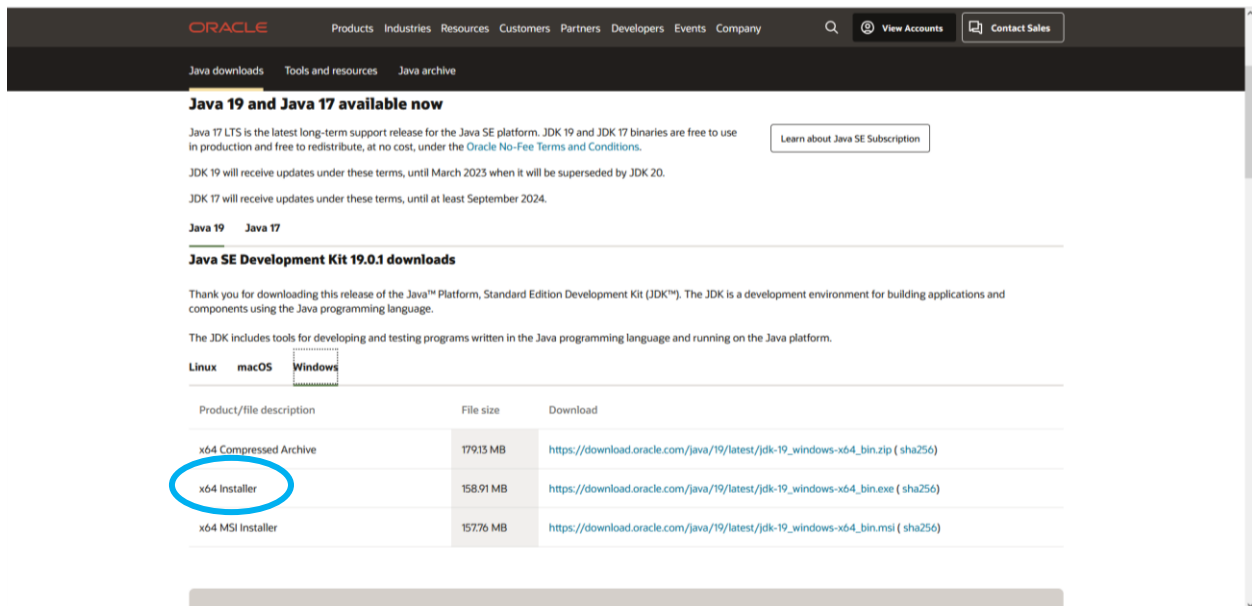


# Download and Install Java Development Kit (JDK) on Windows

## Step 1: Download and Install Oracle Java Development Kit (JDK)

- Download JDK from the official Oracle website:  
<https://www.oracle.com/java/technologies/downloads/#jdk19-windows>
- Verify system specifications to choose appropriate version. Preferably, download the latest version. For Windows, download the latest **x64 Installer of Java SE Development Kit 19.0.1**



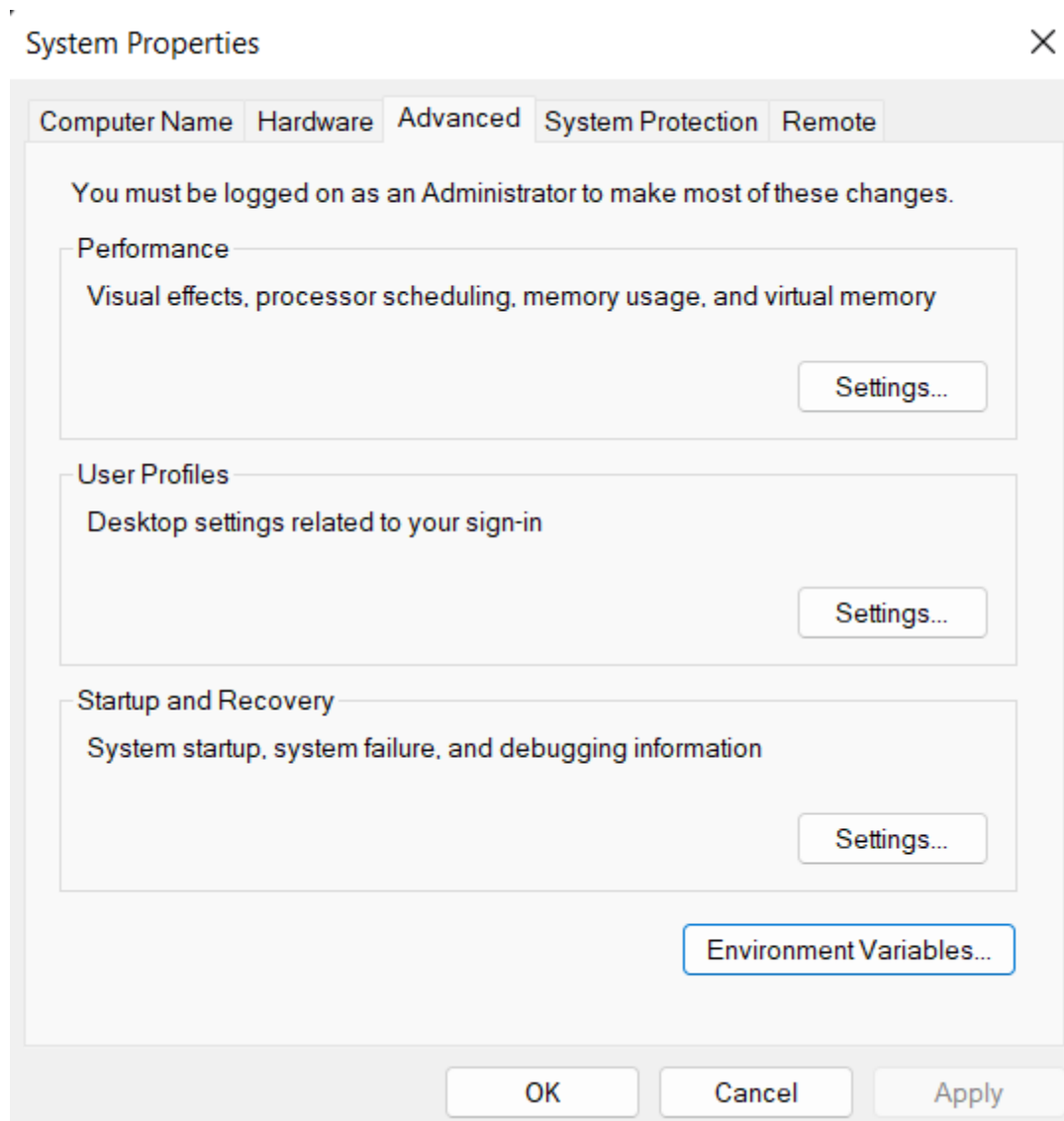
- After the download is complete, proceed to install the JDK by following the steps. Yours should be **Java SE Development Kit 19.0.1**



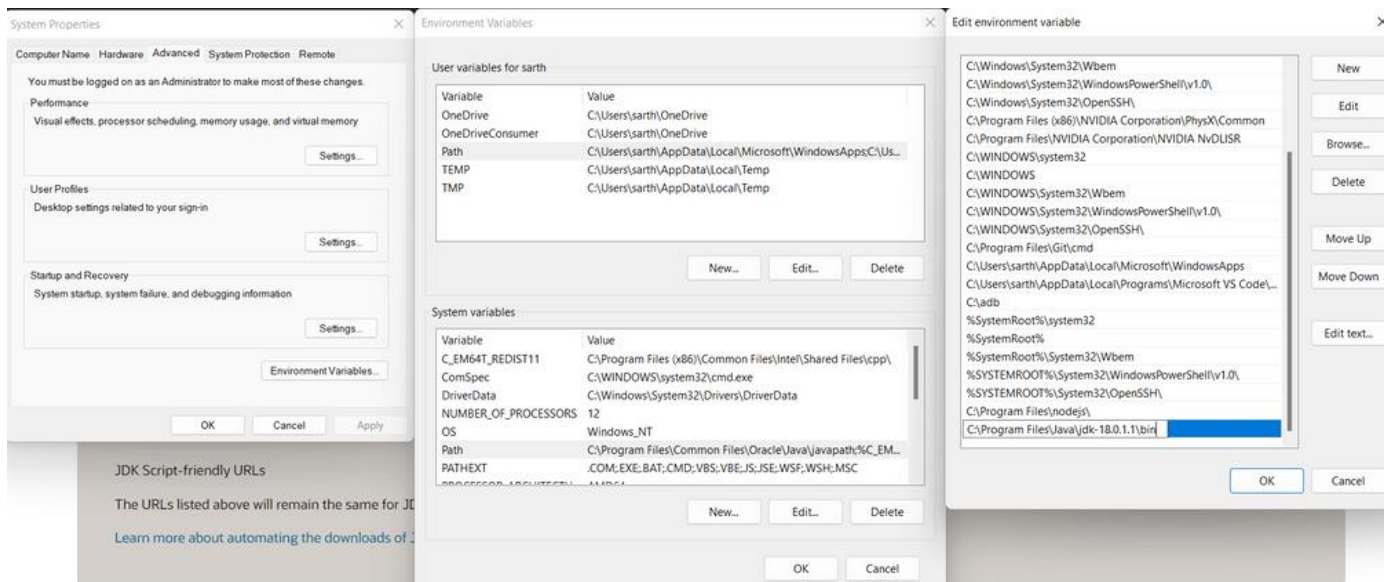
## Step 2: Configure Environment Variables

After the installation is complete, you have to configure environment variables to notify the system about the directory in which JDK files are located.

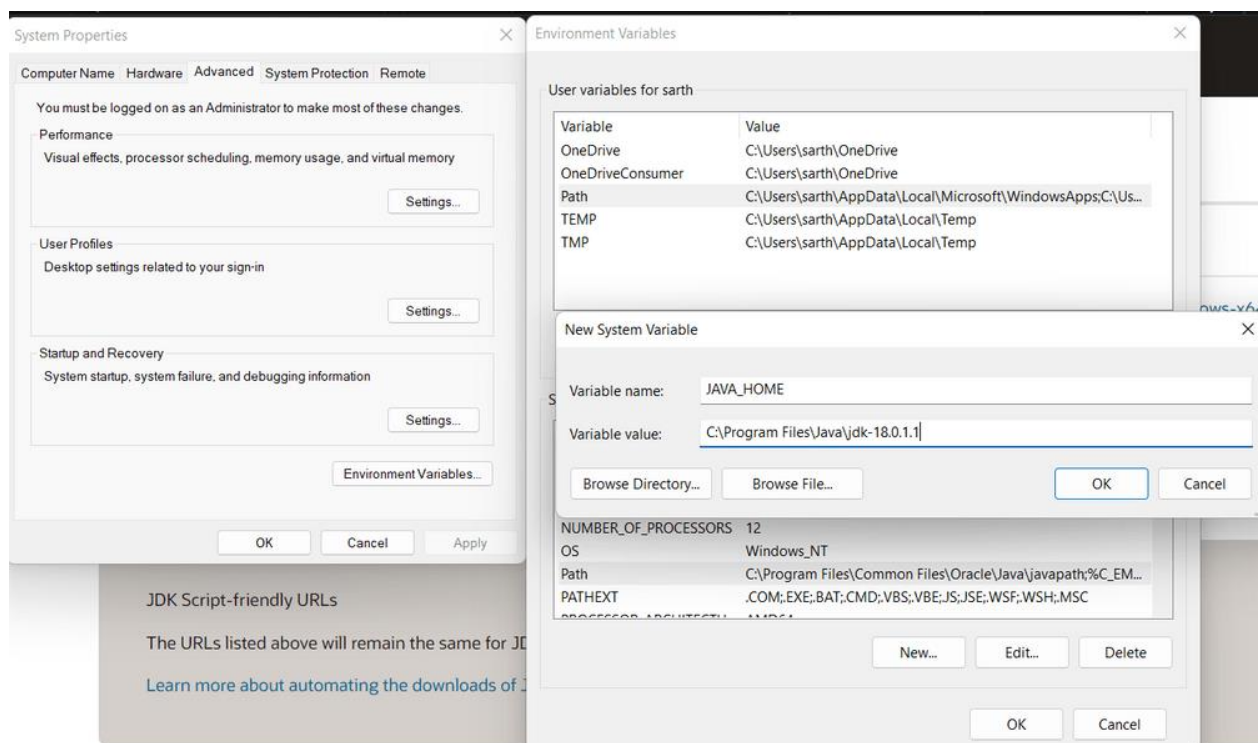
- a. To set the Environment Variables, you need to search Environment Variables in the Task Bar and click on **“Edit the system environment variables”**.



- b. Under the **Advanced** section, Click on **“Environment Variables”**. Under **System variables**, select the **“Path”** variable and click on **“Edit”**. Click on **“New”** then paste the Path Address i.e. **C:\Program Files\Java\jdk-19\bin**. Click on **“OK”**.



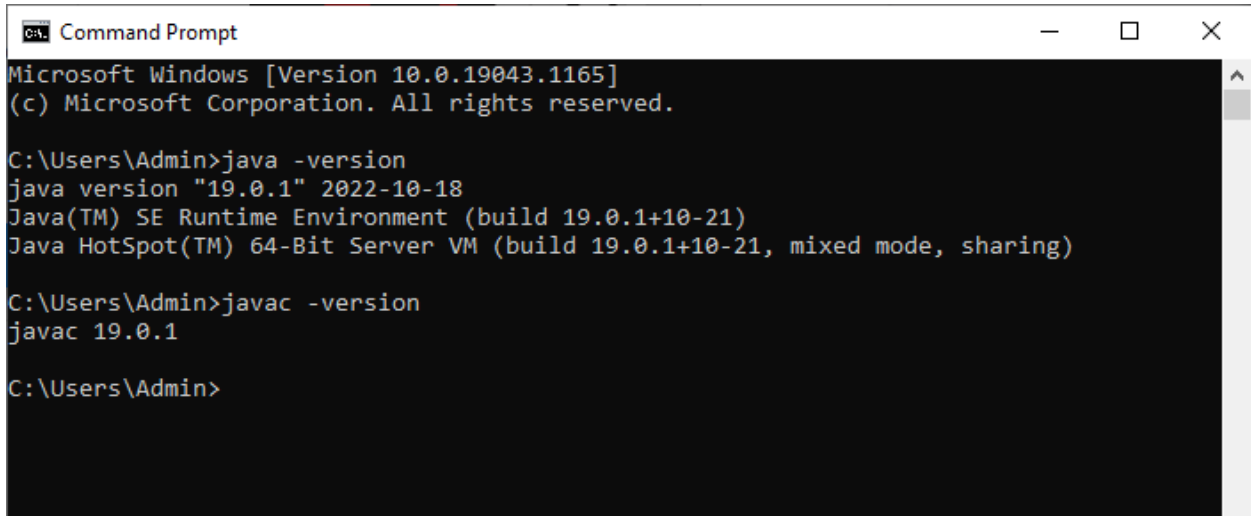
- c. Now, in the **Environment Variables** dialogue, under **System variables**, click on **“New”** and then under **Variable name: JAVA\_HOME** and **Variable value:** paste address i.e. **C:\Program Files\Java\jdk-19** Click on **OK** => **OK** => **OK**.



### Step 3: Check the Java Version

- a. Open **Command Prompt** and enter the following commands:

```
java -version  
javac -version
```



```
Command Prompt  
Microsoft Windows [Version 10.0.19043.1165]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\Admin>java -version  
java version "19.0.1" 2022-10-18  
Java(TM) SE Runtime Environment (build 19.0.1+10-21)  
Java HotSpot(TM) 64-Bit Server VM (build 19.0.1+10-21, mixed mode, sharing)  
  
C:\Users\Admin>javac -version  
javac 19.0.1  
  
C:\Users\Admin>
```

If the commands are unrecognized or you received error messages, kindly go through the steps again. Make sure that the **appropriate path** is provided. The version will also depend on the one you downloaded, it will be version 19.0.1 if you downloaded and installed the latest version from the website.

## Creating your first code in Java.

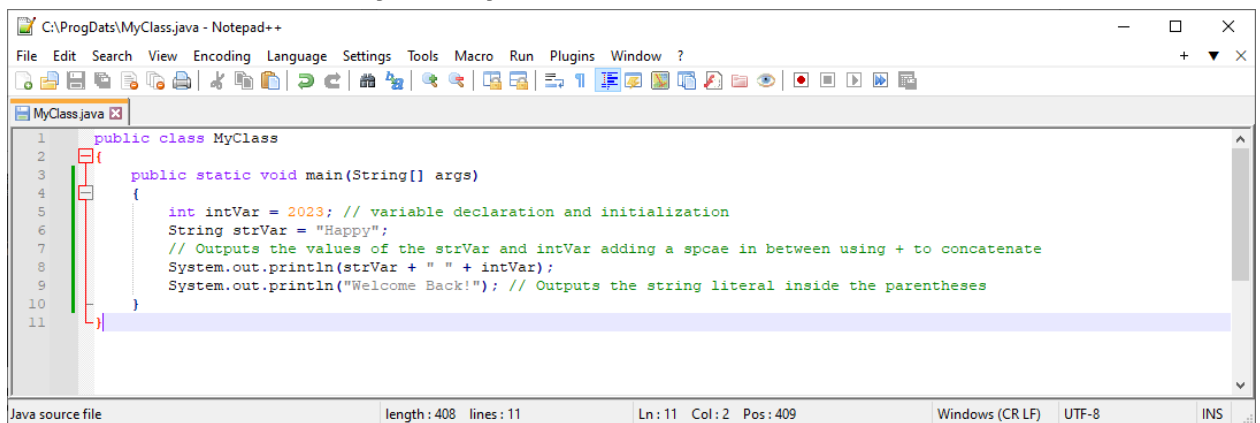
To use the installed JDK, you will save, compile and run a Java program.

### Step 1: Saving a Java Class

- Open Notepad or Notepad++. If you have an existing IDE for Java installed on your device, you may opt to use it.
- Copy the following code:

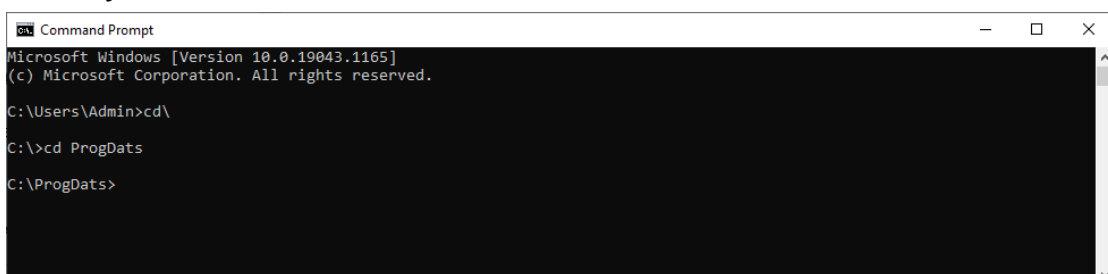
```
public class MyClass
{
    public static void main(String[] args)
    {
        int intVar = 2023; // variable declaration and initialization
        String strVar = "Happy";
        // Outputs the values of the strVar and intVar adding a space in
        // between using + symbol to concatenate
        System.out.println(strVar + " " + intVar);
        // Outputs the string literal inside the parentheses
        System.out.println("Welcome Back!");
    }
}
```

- Create a folder in your drive, name it as **ProgDats**. Save the file in the created folder. Name the file as **MyClass.java**

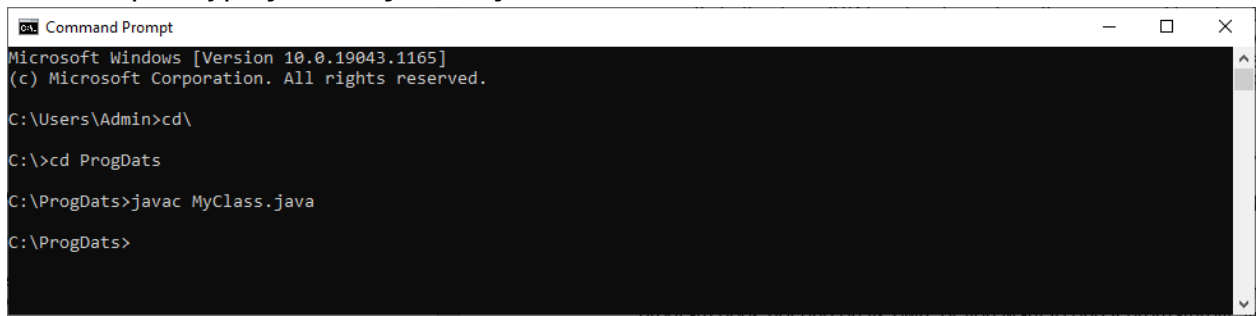


### Step 2: Compiling and Executing a Java program.

- At the command prompt, change from the default drive to the drive where you saved your Java Class.



b. To compile, type **javac MyClass.java**



```
Command Prompt
Microsoft Windows [Version 10.0.19043.1165]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>cd\

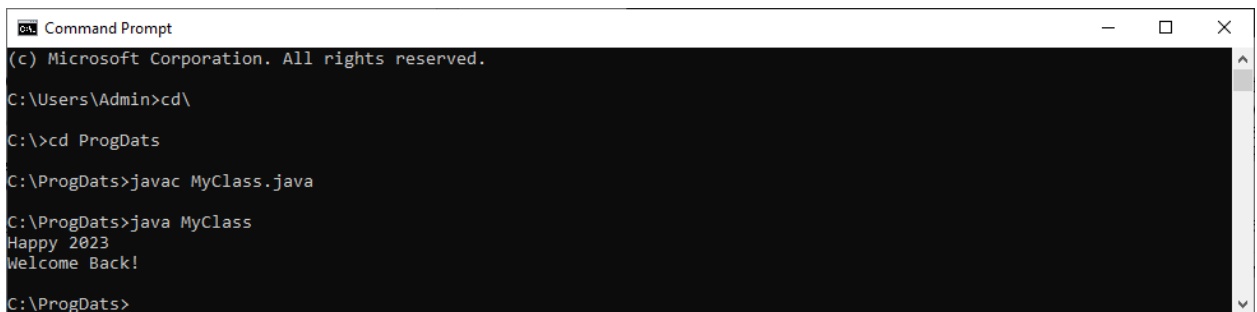
C:\>cd ProgDats

C:\ProgDats>javac MyClass.java

C:\ProgDats>
```

\* If the application doesn't compile successfully, the path might not be set correctly to the Java JDK bin directory where the javac.exe file is located. Also, you might have failed to use the same spelling as the Java filename. When you compile a .java file correctly, the Java compiler creates a .class file that has the same filename as the .java file. Thus, a successful compilation of the First.java file creates a file named First.class.

c. To run a Java application, you use the java command and the class name without the .class extension. For example, after an application named MyClass.java is compiled, producing MyClass.class, you execute the program using the following command: **java MyClass**



```
Command Prompt
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>cd\

C:\>cd ProgDats

C:\ProgDats>javac MyClass.java

C:\ProgDats>java MyClass
Happy 2023
Welcome Back!

C:\ProgDats>
```

\*After the program executes, control is returned to the command prompt. If a program does not end on its own, or you want to end it prematurely, you can press Ctrl+C to return to the command prompt.

Note: Ensure that you have exactly the same code and all steps were followed correctly to execute the output. The details of the program code will be discussed further in class.